

Payoff scale reshapes how language models play the prisoner’s dilemma

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Abstract

Multiplying every payoff in a game by a positive constant leaves preferences, best replies, equilibria and replicator dynamics untouched, so for an agent playing the game the predicted effect of that rescaling is zero by theorem rather than by convention. We use it as an instrument. Five frontier language models played an iterated prisoner’s dilemma across ten payoff scales spanning five orders of magnitude, crossed with five prompt languages and two persona instructions. The grid is complete and exactly balanced: 10,000 dyads, 20,000 agent-games and 200,000 decisions. Cooperation moves in every model, by up to 0.243 [0.200, 0.290], and the movement has a different shape in each one, so the average over models reports a 0.048 swing where the typical model shows 0.136. In a variance decomposition the scale has no significant main effect: 0.7% of cell variance, $p = 0.36$. Its interactions with model (6.5%) and with prompt language (3.8%) are both significant, which is what a pooled benchmark is built to miss. The scale also gates the persona instruction. Where every payoff printed is at most 1, the effect of telling a model it is cooperative rather than selfish reverses sign in three models of five, shifting by 0.680 [0.625, 0.736] in the largest case. Reading the transcripts with a hybrid rule-base and LSTM classifier shows the change is not only in how often the models cooperate but in what they play. The share of agent-games matched exactly by a canonical memory-one rule falls from 31.4% to 19.5% as the payoffs grow, and among the labels the classifier assigns, the mix moves from unconditional towards reciprocal strategies. The effect is already present on the opening move, before any history exists, in four models of five. A cooperation rate reported for a language model is a statement about one arbitrary choice of units, and should be published with its scale sensitivity.

Keywords: machine behaviour; large language models; iterated prisoner’s dilemma; payoff scaling; utility invariance; benchmark validity

1 Introduction

Language models are increasingly deployed as agents that negotiate, coordinate and decide on behalf of their users [1, 2], and a fast-growing literature measures how they behave when they do. The instrument of choice is the classical game, and the prisoner’s dilemma above all, because its strategy space is mapped and its predictions are sharp [3–7]. Studies in this vein report cooperation rates, departures from Nash play, and model-specific signatures [8–12], and those numbers increasingly feed models of mixed human-machine populations [13–15].

Every one of those measurements is made at a particular payoff matrix. The matrix is written down once, in whatever units the experimenter chose, and the cooperation rate is then reported as a property of the model. The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, and such a utility is defined only up to a positive affine transformation [16, 17]. Writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no replicator trajectory. Under the theory the two matrices are not two games written differently. They are one game.

This equivalence makes a positive rescaling the sharpest available probe of whether a language model is playing the game it was shown. Most manipulations one might apply to a game-theoretic prompt have arguable effects: a change of framing carries new words [18, 19], a change of persona carries new information about the agent [20], a change of language carries a different training distribution. A rescaling carries none of these. It changes the numerals and nothing else. If behaviour is a function of the game, the effect is exactly zero, and that null needs no defending because it is a theorem. If behaviour moves, then whatever the model is responding to is not the game.

Sensitivity to details that should not matter is already documented for language models in other settings. Accuracy shifts with the order of in-context examples [21] and with semantically equivalent rephrasings of a prompt [22]. Those findings establish that surface form matters; they do not settle what happens inside a game, because reordering examples or rewording an instruction has no theoretical guarantee attached. Payoff rescaling does, and the guarantee is what makes the measurement interpretable: any movement is a measured departure from a stated invariance, not a difference between two conditions whose expected gap was unknown to begin with.

This paper takes the rescaling as its object of study, and asks four questions of it. Does strategic behaviour survive multiplying every payoff by a positive constant, and by how much does it fail to? Does the answer depend on the language the game is posed in, and on the persona the model is told to adopt? Does the rescaling change only how often a model cooperates, or the strategy it is playing? And where in the game does the effect enter: at the opening move, before any history exists, or through the course of play?

We answer them on a complete and exactly balanced grid: five frontier models, ten payoff scales from $\lambda = 0.01$ to $\lambda = 10^3$, five prompt languages, two persona instructions in all four pairings, ten replicates and ten rounds. The corpus holds 10,000 dyads, 20,000 agent-games and 200,000 recorded decisions. Four findings follow. First, cooperation moves in all five models, by up to 0.243, and the movement is significant against a permutation null that destroys only the association with the scale. Second, the shape of the movement differs between models to the point of anticorrelation, so pooling reports the effect as roughly a third of its typical within-model size. In a variance decomposition the scale main effect is not significant while its interactions are. Third, the scale gates the persona instruction, reversing its sign in three models of five below the point where every printed payoff falls to 1 or less. Fourth, the strategy itself changes: play becomes measurably less rule-governed as the payoffs grow, and the composition moves from unconditional towards reciprocal rules. The effect is already visible on the opening move.

We should be explicit about what this design can and cannot establish. It measures departures from an invariance across a broad but finite grid; it does not identify the mechanism inside the model that produces them, and with five models it cannot support claims about model families

or scale in parameters. The corpus is one game, one prompt template family and one horizon, so the generality of the specific shapes reported here is untested. What the design does establish is that the departures exist, are large relative to their sampling variability, and are structured rather than noisy.

2 Results

2.1 Design

Both agents in each dyad are given the standard prisoner’s-dilemma prompt of the FAIRGAME protocol [23], which states the payoffs as years of imprisonment and instructs the agent to minimise them. At $\lambda = 1$ the matrix the agent is shown is $T = 0$, $R = 2$, $P = 6$, $S = 10$ in those penalty units, so that the mutually cooperative cell carries the lowest symmetric penalty and the unilateral defection cell carries none at all. Under this framing OptionA is the dominant, defecting action and OptionB is the cooperative one, and cooperation rates below are computed accordingly. The manipulation multiplies all four cells by λ , taking ten values from 0.01 to 10^3 (Fig. 1a). Nothing else in the prompt changes.

The grid is crossed with five prompt languages (English, French, Vietnamese, Chinese, Arabic), with a persona instruction that names each agent as cooperative or selfish in all four pairings, and with ten replicates. Each game runs ten rounds, with the horizon announced. Every one of the 250 cells of the design contains exactly 200 dyads; the build refuses to proceed otherwise, because a silently truncated cell is a failure this kind of corpus is prone to. Overall cooperation is 0.565.

2.2 Cooperation moves under a transformation that cannot matter

Cooperation depends on the payoff scale in all five models (Fig. 1b, Table 1). The range across the ten scales runs from 0.072 [0.060, 0.133] in Gemini 3.1 Flash-Lite to 0.243 [0.200, 0.290] in GPT-5.4 Nano, with Gemini 3.5 Flash-Lite at 0.191 [0.165, 0.240]. Confidence intervals resample whole dyads rather than rounds, because the two agents of a dyad play each other and the ten rounds within an agent-game are strongly autocorrelated.

Because the null here is a theorem, the appropriate test destroys the association with the scale and leaves everything else in place. Shuffling the scale label within language and persona, and recomputing the range 2,000 times, puts the observed value beyond every permuted one in all five models ($p = 0.0005$, the smallest value 2,000 draws can return). We also fit a per-model logistic regression of the round-level decision on the scale as a ten-level factor, with language and both personas as covariates and standard errors clustered on the dyad. It rejects the joint null on the nine scale contrasts in every model (χ^2_9 between 32.3 and 164.1, all $p \leq 2 \times 10^{-4}$). The largest contrast against $\lambda = 1$ ranges from 0.36 to 0.74 in log-odds.

The shapes are unlike one another. GPT-5.4 Nano rises steeply from 0.438 at $\lambda = 0.01$ and saturates near 0.65; Gemini 3.1 Flash-Lite drifts downward across the whole range; Claude Haiku 4.5 dips to a minimum at $\lambda = 0.25$ and recovers; Gemini 3.5 Flash-Lite oscillates within a high band. Only the direction of the smallest scales is at all shared, and even that is not universal.

one game, three renderings

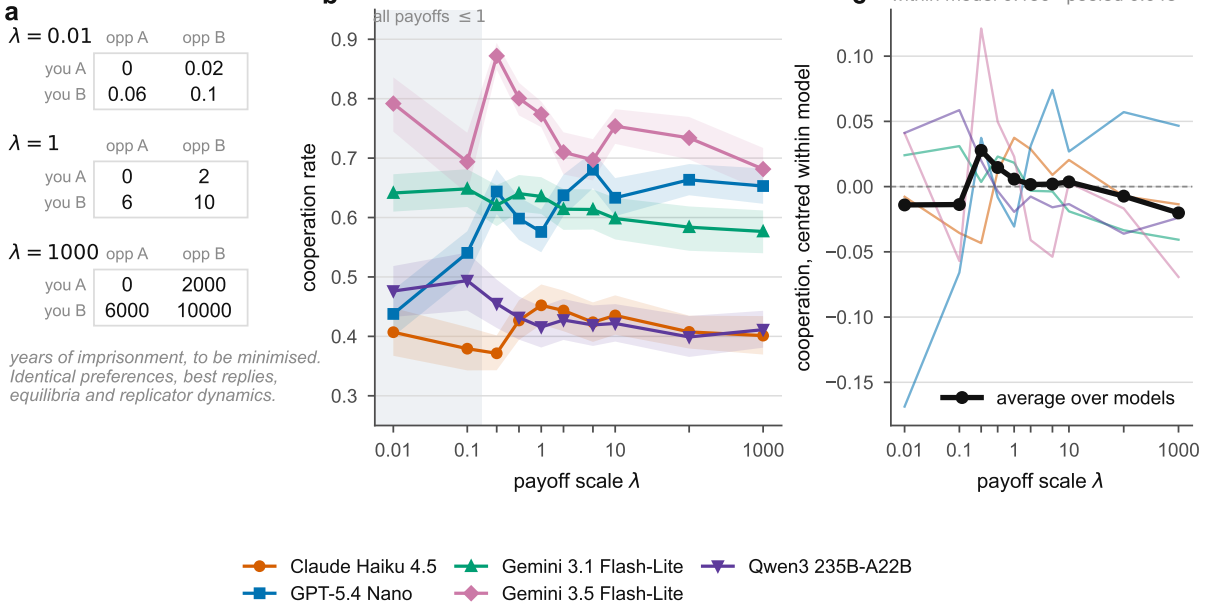


Figure 1: A rescaling that cannot matter, and does. (a) The same prisoner’s dilemma at three payoff scales. The agent is shown years of imprisonment and told to minimise them; multiplying every cell by λ leaves preferences, best replies, equilibria and replicator dynamics unchanged. (b) Cooperation against the payoff scale for each model, with 95% intervals from a bootstrap that resamples whole dyads. The shaded band marks the two scales at which every payoff printed is at most 1. Each point aggregates 200 dyads. (c) The same curves centred within model, with their average in black. The models disagree in shape to the point of anticorrelation, so the average swings 0.048 where the typical model swings 0.136.

2.3 Averaging across models reports the effect as almost absent

The disagreement in shape has a direct consequence for how such measurements are usually reported. Averaging the five model curves gives a pooled curve whose range is 0.048, while the mean within-model range is 0.136, an attenuation of 2.8 times (Fig. 1c). The cause is visible in the pairwise correlations between model curves: they average -0.10 and run from -0.72 to $+0.73$, with six of the ten pairs negative. Curves that move in opposite directions cancel when added.

A variance decomposition on the 250 cell means makes the same point in the language a benchmark report would use. The model accounts for 56.5% of the variance across cells and the model-by-language interaction for 19.7%. The payoff scale as a main effect accounts for 0.7% and is not significant ($F_{9,144} = 1.10$, $p = 0.36$). Its interactions are: model by scale carries 6.5% ($p < 10^{-3}$) and language by scale 3.8% ($p = 0.041$). A study that fits a main effect for the payoff scale, pooling models and languages, will therefore conclude correctly that there is no average effect, and will be wrong about every individual model in the pool.

2.4 The prompt language sets how large the effect is

The prompt language changes both the level of cooperation and its sensitivity to the scale (Fig. 2). Level differences are large and model-specific, which is why the model-by-language interaction dominates the decomposition above: Claude Haiku 4.5 cooperates 0.574 in Chinese and 0.234 in Vietnamese, while GPT-5.4 Nano cooperates 0.798 in English and 0.458 in Arabic.

Table 1: Cooperation moves with the payoff scale in every model. Minimum and maximum cooperation rate over the ten scales, their range, a 95% confidence interval on that range from a bootstrap that resamples whole dyads, and a permutation test that shuffles the scale label within language and persona (2,000 draws, so 0.0005 is the smallest attainable value). The last two columns give a Wald test on the nine scale contrasts of a per-model logistic regression of the round-level decision, with standard errors clustered on the dyad ($n = 40,000$ decisions and 2,000 clusters per model).

Model	cooperation			95% CI	p_{perm}	Wald test	
	min	max	range			χ^2_9	p
Claude Haiku 4.5	0.371	0.452	0.081	[0.062, 0.135]	< 0.001	32.3	< 0.001
GPT-5.4 Nano	0.438	0.680	0.243	[0.200, 0.290]	< 0.001	164.1	< 0.001
Gemini 3.1 Flash-Lite	0.577	0.648	0.072	[0.060, 0.133]	< 0.001	83.1	< 0.001
Gemini 3.5 Flash-Lite	0.682	0.872	0.190	[0.165, 0.240]	< 0.001	129.0	< 0.001
Qwen3 235B-A22B	0.399	0.494	0.095	[0.068, 0.158]	< 0.001	96.9	< 0.001

Table 2: The payoff scale gates the persona instruction. The persona effect is the cooperation rate of an agent told it is cooperative minus that of an agent told it is selfish. It is reported separately for the two scales at which every payoff printed is at most 1 ($\lambda \leq 0.1$) and for the eight at which some payoff exceeds it. The shift between the two regimes carries a 95% confidence interval from the dyad bootstrap; the last column records whether the persona effect changes sign.

Model	persona effect		shift	95% CI	sign flip
	$\lambda \leq 0.1$	$\lambda \geq 0.25$			
Claude Haiku 4.5	−0.222	+0.207	0.429	[0.389, 0.471]	yes
GPT-5.4 Nano	−0.135	+0.018	0.153	[0.105, 0.200]	yes
Gemini 3.1 Flash-Lite	+0.577	+0.680	0.103	[0.075, 0.131]	no
Gemini 3.5 Flash-Lite	−0.456	+0.224	0.680	[0.625, 0.736]	yes
Qwen3 235B-A22B	−0.911	−0.679	0.233	[0.210, 0.258]	no

The sensitivity differences are the new observation. Within a single model, the range of cooperation across the ten scales itself depends on the language: 0.136 in English against 0.314 in French for Claude Haiku 4.5, and 0.162 in Chinese against 0.412 in English for Gemini 3.5 Flash-Lite. Across all model-language pairs the range runs from 0.080 to 0.449. The scale sensitivity of a model is therefore not a single number even for that model; it is a property of the model and the language together, and the language-by-scale interaction in the decomposition is significant.

2.5 The payoff scale gates the persona instruction

Each agent is told it is either a cooperative or a selfish player, and the difference between the cooperation rates of those two instructions measures how much the persona takes hold. That difference depends on the payoff scale, and the dependence is not gradual (Fig. 3, Table 2).

The natural boundary in this design is where the printed numbers stop being sub-unit. With base payoffs 0, 2, 6, 10, the largest number shown is 10λ , so every payoff is at most 1 exactly when $\lambda \leq 0.1$, which two of the ten scales satisfy. Across that boundary the persona effect reverses sign in three models of five. In Gemini 3.5 Flash-Lite it runs from −0.456 below to +0.224 above, a shift of 0.680 [0.625, 0.736]. In Claude Haiku 4.5 it runs from −0.222 to +0.207,

Table 3: Larger payoffs make play less rule-governed in three models of five. The left block gives the share of agent-games matched exactly by one of the four canonical rules, below and above the sub-unit boundary, and the coefficient of a logistic regression of that indicator on $\log_{10} \lambda$. The right block gives the mean number of rounds that violate the nearest canonical rule at the smallest and largest scale, and the same trend fitted by least squares. Both use standard errors clustered on the dyad. Neither quantity depends on the learned classifier.

Model	exactly one rule fits (%)				rounds violating the nearest rule			
	$\lambda \leq 0.1$	$\lambda \geq 0.25$	β	p	$\lambda=0.01$	$\lambda=10^3$	β	p
Claude Haiku 4.5	17.9	3.5	-0.545	< 0.001	1.70	2.02	+0.073	< 0.001
GPT-5.4 Nano	9.8	7.6	-0.038	0.441	2.02	2.19	+0.038	0.052
Gemini 3.1 Flash-Lite	17.8	26.3	+0.125	< 0.001	1.08	1.03	-0.019	0.262
Gemini 3.5 Flash-Lite	32.2	15.6	-0.242	< 0.001	0.48	1.36	+0.194	< 0.001
Qwen3 235B-A22B	72.1	53.6	-0.267	< 0.001	0.27	1.14	+0.202	< 0.001

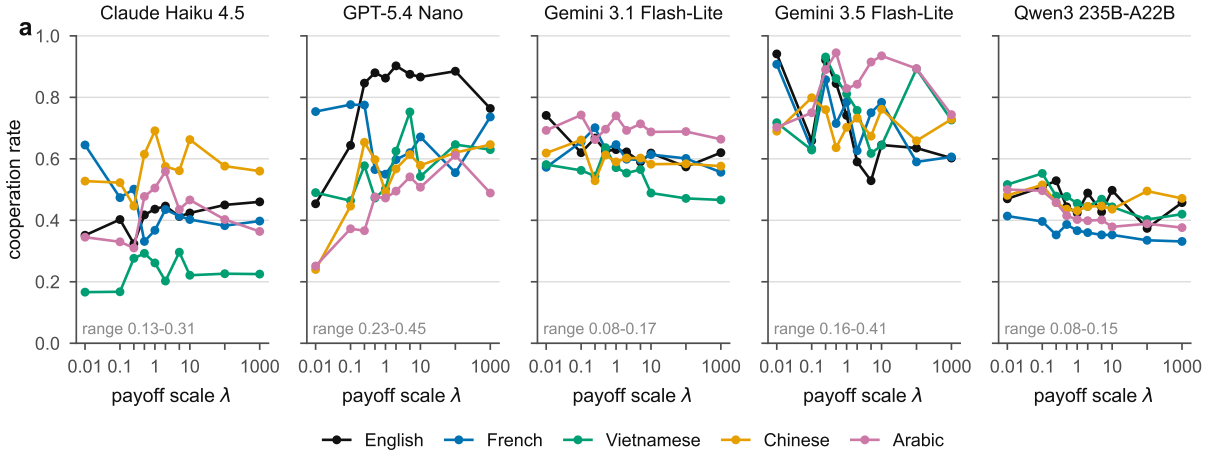


Figure 2: The prompt language sets how large the scale effect is. Cooperation against the payoff scale, separately for each model and each of the five prompt languages; 40 dyads per point. The annotation in each panel gives the smallest and largest across-scale range among that model’s five languages. The same model can be nearly flat in one language and swing by more than 0.4 in another.

a shift of 0.429 [0.389, 0.471], and in GPT-5.4 Nano from -0.135 to $+0.018$, a shift of 0.153 [0.105, 0.200]. Below the boundary these models cooperate *more* when told they are selfish. The two models that do not reverse still move in the same direction, by 0.103 [0.075, 0.131] and 0.233 [0.210, 0.258].

The boundary is a fact about magnitude and not about decimal notation, and the grid separates the two. At $\lambda = 0.25$ the payoffs are 0, 0.5, 1.5, 2.5: still written as decimals, but no longer all below 1. The persona effect at that scale already has its supra-unit sign and size in all three reversing models. Whatever the models are responding to, it tracks how large the numbers are rather than whether a decimal point appears.

Qwen3 235B-A22B deserves separate comment because it inverts the persona instruction at every scale, cooperating 0.072 under the cooperative persona and 0.798 under the selfish one. The prisoner’s dilemma story supports two readings of what it means to cooperate, with one’s partner or with the authorities, and this model appears to resolve that ambiguity opposite to the convention the payoff matrix encodes. The inversion is not our subject, but its magnitude

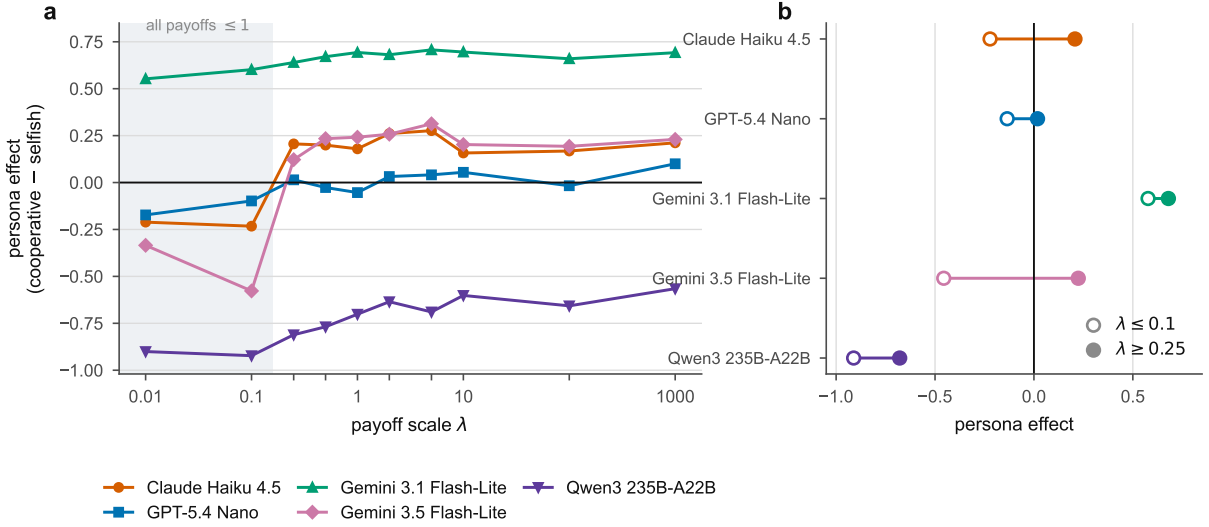


Figure 3: The payoff scale gates the persona instruction. (a) Persona effect, the cooperation rate under a cooperative persona minus that under a selfish one, against the payoff scale. The shaded band marks the two scales at which every payoff printed is at most 1. Three models of five cross zero at that boundary. (b) The same contrast summarised: open markers give the mean persona effect at $\lambda \leq 0.1$, filled markers at $\lambda \geq 0.25$. Every model shifts in the same direction, and the shift excludes zero in all five (Table 2).

is scale-dependent in the same way as everything else here: it attenuates monotonically from -0.912 to -0.679 as the payoffs grow.

2.6 The scale changes which strategy is played

A cooperation rate compresses a ten-round trajectory to one number, and two models with the same rate can be playing different games. We therefore read each agent-game against the four canonical memory-one strategies: AllC, AllD, Tit-for-Tat and Win-Stay-Lose-Shift. The procedure is hybrid, with a rule base doing the deducing and a learned sequence model only ranking what the rules leave open (§4). The distinction matters for how the results should be read, so the read-out reports three provenances separately: the agent-games where exactly one rule fits the trajectory exactly, where several fit and no method could separate them, and where none fits.

Play becomes less rule-governed as the payoffs grow (Fig. 4a, Table 3). Pooling models, the share matched exactly by one canonical rule falls from 31.4% at $\lambda = 0.01$ to 19.5% at $\lambda = 10^3$, and the share matched by none rises from 54.4% to 69.2%. Per model the trend is significant and negative in Claude Haiku 4.5 ($\beta = -0.545$ per decade of λ), Qwen3 235B-A22B (-0.267) and Gemini 3.5 Flash-Lite (-0.242). It is significant and positive in Gemini 3.1 Flash-Lite ($+0.125$), and absent in GPT-5.4 Nano ($p = 0.44$). The heterogeneity is the same one seen in the cooperation rates, and it would again cancel under pooling.

The continuous version of the same measurement needs no labels at all: the number of rounds that violate the nearest canonical rule (Fig. 4b). It rises from 0.48 to 1.37 in Gemini 3.5 Flash-Lite and from 0.27 to 1.14 in Qwen3 235B-A22B between the smallest and largest scale, with significant positive trends there and in Claude Haiku 4.5. Because this quantity is computed by exact rule matching, it does not depend on the classifier at all, which is why we

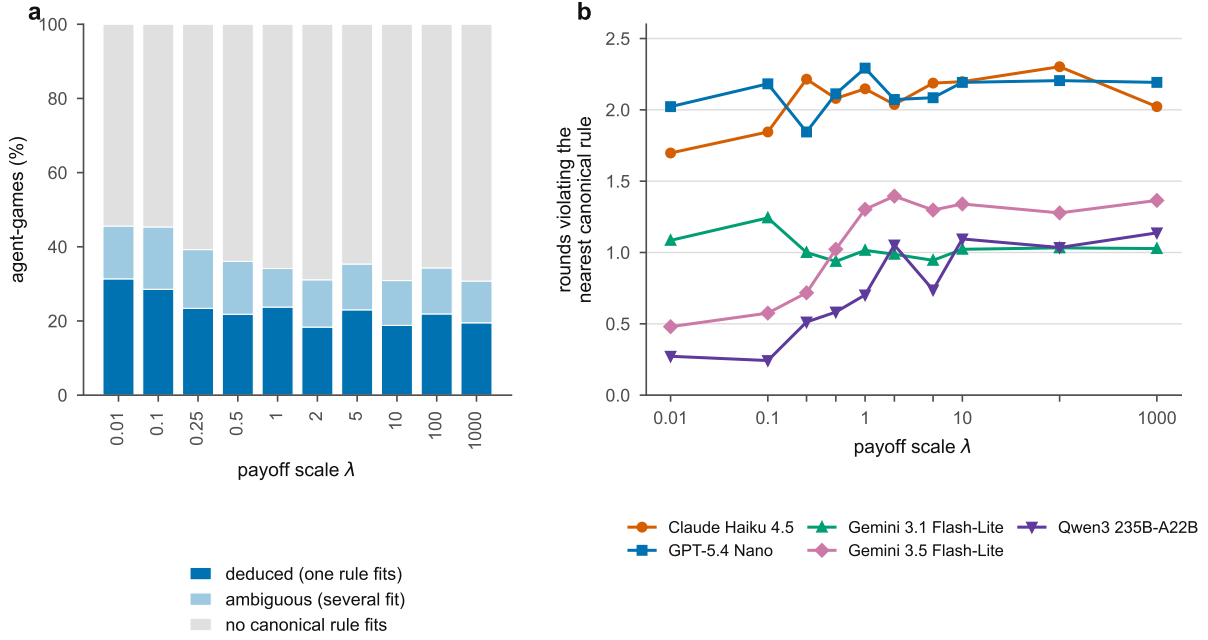


Figure 4: Larger payoffs change what is played, not only how often it cooperates. (a) Provenance of the strategy read-out against the payoff scale, pooled over models. Exact rule matches become rarer as the payoffs grow. **(b)** Mean number of rounds violating the nearest canonical rule. This quantity is computed by exact matching and involves no learned component.

lead the strategy analysis with it.

The composition shifts too, and in a consistent direction: away from unconditional and towards reciprocal play (Fig. 5). Pooled across models, AllC falls from 45.6% to 40.4% of agent-games and Tit-for-Tat rises from 9.2% to 13.0% between the extreme scales. The per-model movements are larger and again not uniform. GPT-5.4 Nano moves from 29.8% to 50.8% AllC while its AllD share falls from 46.0% to 17.5%. Gemini 3.5 Flash-Lite moves the other way on AllC, from 73.8% to 51.5%, with Tit-for-Tat roughly doubling from 6.8% to 14.3%.

This last result is the one in the paper that leans hardest on the learned component, and the provenance is not uniform across the four labels. Of the agent-games carrying the AllC label, 26.0% were deduced from an exact rule match and a further 29.6% came from a rule set the LSTM ranked within; for AllD the deduced share is 32.7%. For Tit-for-Tat and Win-Stay-Lose-Shift, however, 95.4% and 97.3% of the labels are attributions the LSTM makes on trajectories that match no canonical rule at all. A movement in the Tit-for-Tat share is therefore a statement about which canonical rule the play most resembles, not a claim that the model is running Tit-for-Tat, and we do not read it as one.

2.7 The effect is present on the opening move

A rescaling could act through the course of play, by changing how a model reads its opponent’s history, or it could act on the reading of the matrix itself. The opening move separates the two, because at round 1 no history exists and nothing but the matrix and the instructions has been seen.

The effect is there already (Fig. 6). In four models of five the range of opening-move cooperation across the ten scales exceeds the range over rounds 2 to 10. The factor is 3.7 in

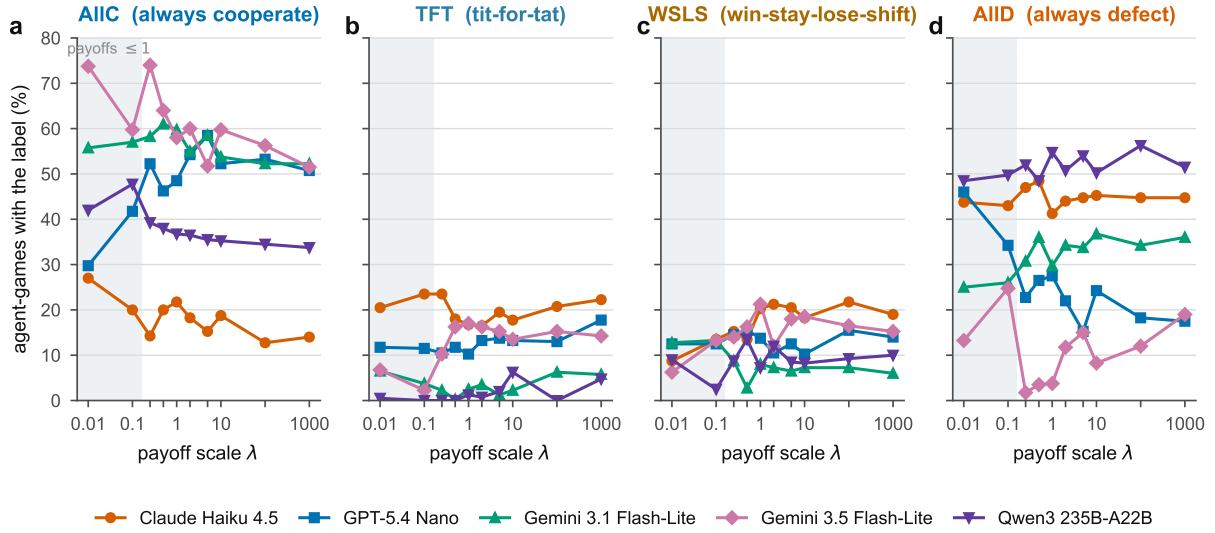


Figure 5: Which strategy is played, against the payoff scale. One panel per canonical rule: (a) AllC, (b) Tit-for-Tat, (c) Win-Stay-Lose-Shift, (d) AllD. Each point is the share of that model’s 400 agent-games at that scale carrying the label, so the four panels sum to 100% within every model and scale and share a common vertical axis. Shaded bands mark the two scales at which every payoff printed is at most 1. The models do not move together: GPT-5.4 Nano trades AllD for AllC as the payoffs grow while Gemini 3.5 Flash-Lite moves the opposite way on AllC, and Claude Haiku 4.5 shifts from AllC towards Win-Stay-Lose-Shift. The AllC and AllD panels rest partly on exact rule matching, whereas the Tit-for-Tat and Win-Stay-Lose-Shift labels are 95.4% and 97.3% attributions made where no canonical rule fits, and should be read as a resemblance rather than as an identification.

Gemini 3.1 Flash-Lite, 2.0 in Claude Haiku 4.5, 1.9 in Gemini 3.5 Flash-Lite and 1.3 in GPT-5.4 Nano. The exception is Qwen3 235B-A22B, whose opening move is almost entirely determined by its persona and therefore sits at 0.50 by construction at nearly every scale, leaving little room to vary.

The reading we take from this is deflationary about dynamics. Whatever the payoff scale does, it does most of it before the game has begun. That timing is consistent with the models forming an initial disposition by reading the magnitudes off the matrix, and inconsistent with an account in which the scale changes how history is weighed.

3 Discussion

The measurement reported here is unusual in having a null that is a theorem. That property is what lets a movement of 0.243 in cooperation be read as a departure from invariance rather than as the difference between two conditions. All five models depart, and none is close to invariant. The departures are structured: they enter at the opening move, they scale with the prompt language, they reverse the persona instruction below a magnitude boundary, and they change which canonical strategy the transcript matches.

The practical consequence is for how such numbers are reported. A cooperation rate published for a language model is currently understood as a property of the model. On this evidence it is a property of the model, the units the matrix was written in, and the language it was posed in, jointly. Two laboratories running the same protocol with matrices differing by a constant factor would report cooperation rates differing by up to 0.24 and would have no way to tell,

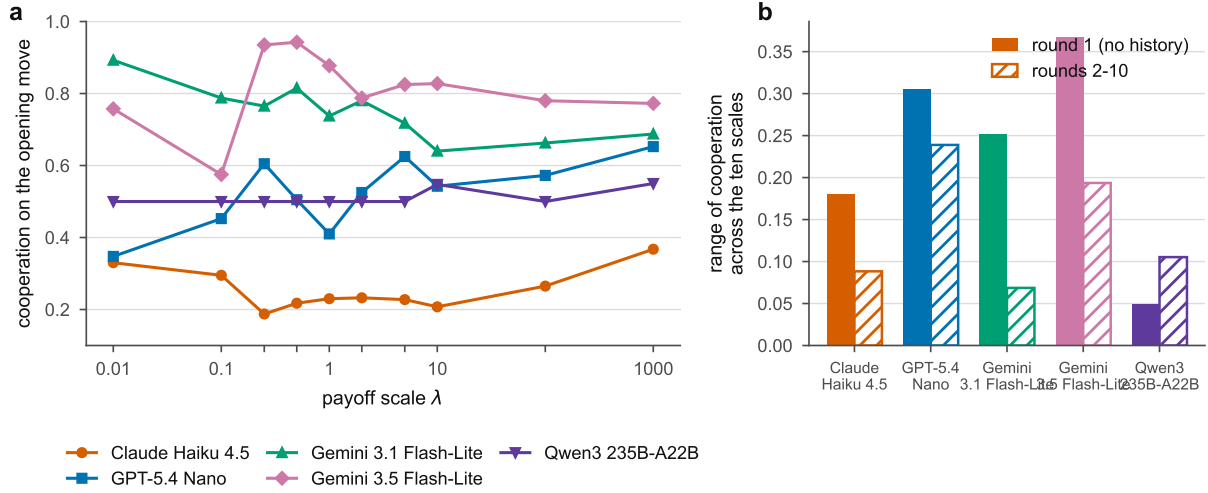


Figure 6: The scale effect is present before any history exists. (a) Cooperation on the opening move against the payoff scale. At round 1 the agent has seen the matrix and the instructions and nothing else. (b) Range of cooperation across the ten scales, on the opening move (solid) and over rounds 2 to 10 (hatched). In four models of five the opening move carries the larger range. Qwen3 235B-A22B is the exception: its opening move is fixed by its persona at 0.50 and cannot vary.

from the numbers alone, that they had not disagreed about the model. We would suggest that a scale-sensitivity measurement accompany any published cooperation rate, in the same way an error bar does, and that it be measured per model rather than pooled.

The pooling result deserves emphasis because it is the mechanism by which this effect could have gone unnoticed. The scale has no significant main effect in our data, and a study designed to detect a main effect would have correctly found none while every model in it was moving substantially. Heterogeneous effects that cancel are invisible to pooled designs by construction, and the remedy is not a larger sample but a per-model analysis.

Two observations point beyond benchmark hygiene. The first is that the boundary gating the persona instruction is one of magnitude, not of notation. Sub-unit payoffs behave differently from supra-unit ones, and $\lambda = 0.25$, still printed with a decimal point, already behaves like the large scales. The second is that larger payoffs make play measurably less like any canonical memory-one rule while shifting the mix towards reciprocity. Both are consistent with the models treating the printed magnitudes as carrying information about how much is at stake, which is precisely the information a positive rescaling does not contain.

The limitations are those of the design. Five models cannot support claims about families or about parameter count, and the specific shapes we report are properties of these five. One game, one prompt template family and one horizon of ten rounds leave the generality of the shapes untested, though the consistency of the qualitative findings across five models and five languages argues they are not artefacts of a single configuration. We measure departures from an invariance; we do not identify the mechanism that produces them, and nothing here distinguishes a tokenisation effect on numerals from a learned association between large stakes and caution. Finally, our reference for "large" is the sampling variability of these data, through the dyad bootstrap and the permutation null, rather than an independent repeat of the whole protocol. A measured test-retest floor from re-run sweeps would be a stronger comparison, and it is the obvious next measurement.

The strategy read-out carries its own caveat, which is why its provenance is reported. Only 23.0% of agent-games are matched exactly by a single canonical rule; 13.2% are matched by several at once, which no method can separate, and 63.7% by none. The claims we draw from it are deliberately the ones that survive those numbers. The share of exact matches and the distance to the nearest rule are computed without any learned component; the compositional claim is the weakest of the three and is presented as such. The asymmetry between labels sharpens that caveat rather than softening it: the AllC and AllD labels carry deduced shares of 26.0% and 32.7%, while Tit-for-Tat and Win-Stay-Lose-Shift are 95.4% and 97.3% attributions to trajectories no canonical rule fits. Any reading of the compositional result as evidence that these models implement Tit-for-Tat would be unsupported by our data.

4 Materials and methods

Corpus. Games follow the FAIRGAME protocol [23] for the iterated prisoner’s dilemma without communication, with the number of rounds announced. The design crosses ten payoff scales $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$, five prompt languages, four persona pairings and ten replicates, at ten rounds per game, for the five models named in the figures. Each of the 250 model-by-language-by-scale cells contains exactly 200 dyads. The ingest refuses to write its tables if any cell departs from that, if any game is not ten rounds long, or if the payoff scale recovered from the recorded payoffs disagrees with the one the cell was run at. Decisions are recorded as OptionA or OptionB and mapped to defection and cooperation respectively, following the penalty framing of the prompt.

Uncertainty. The two agents of a dyad play each other, and the ten rounds within an agent-game are autocorrelated. All intervals therefore resample whole dyads with replacement (1,500 draws, percentile method [24]), and all regressions cluster standard errors on the dyad. The permutation test shuffles the payoff-scale label within language and persona, which leaves the design intact and destroys only the association under test, and uses 2,000 draws. The variance decomposition is a type-II analysis of variance on the 250 cell means, where the design is exactly balanced.

Strategy read-out. Each agent-game is encoded as a sequence of tokens pairing the previous round’s joint outcome with the current action, an alphabet shared between real transcripts and a synthetic corpus. A rule base then asks, for each of AllC, AllD, Tit-for-Tat and Win-Stay-Lose-Shift, whether the trajectory is exactly what that rule would have played given the history that actually occurred. The answer is a set rather than a label, because over ten rounds several canonical rules routinely prescribe the same moves. A player who cooperated throughout against a cooperator is exactly consistent with AllC, Tit-for-Tat and Win-Stay-Lose-Shift at once, and reporting all three is correct rather than a failure. A single-layer LSTM [25] (24-dimensional embeddings, 96 hidden units) ranks within that set, and decides alone only when no rule fits. It is trained on synthetic trajectories of the four rules at zero and 5% execution noise, balanced at 40,320 sequences per strategy per level. It reaches 98.5% accuracy on a held-out split, 93.6% at an unseen 10% noise level and 81.1% at 20%. No language-model transcript is used in training.

Availability. Analysis code, the tidy tables and the scripts that generate every figure and table in this paper are in `Analysis/scaling/`; the transcripts are in `Dataset/data_fairgame_frontier_llm/`.

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